



Unit 3 · Outcome 1 · Maintaining Biodiversity

The dividends nature pays:ecosystem services

KK03

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. Ecosystems as a source of renewable services for human health and well-being: provisioning, regulating, supporting and cultural services.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



WHERE THIS SITS

The dot point, in VCAA's exact words

◆ NOTE

Key knowledge “Ecosystems as a source of renewable services for human health and well-being: provisioning, regulating, supporting and cultural services.”

Two ideas are doing the heavy lifting in that sentence. First, renewable — these services regenerate naturally as long as the ecosystem is kept functioning. Second, human health and well-being — every service is framed by how it benefits people, which is why this dot point sits so close to value systems and sustainability later in the chapter.

- Name and define all four categories of ecosystem service
- Classify a real-world example into the correct category
- Explain why supporting services underpin the other three
- Evaluate the consequences when a service is degraded or lost

Ecosystem services connect biodiversity to human value. It feeds directly into later dot points — threats (KK10), management (KK11–12), value systems (KK14) and sustainability (KK15). Markers reward students who can move from “name it” to “so what does losing it cost us?”

THE MENTAL MODEL

Nature's invisible utility company

Imagine a single company that quietly runs your water treatment plant, your power station, your farm, your insurance against floods, and your local park — and never sends an invoice. That's an ecosystem. The four “departments” of that company map exactly onto the four categories of service.

◆ ANALOGY

Carry this analogy through the module Four departments, one free utility Provisioning is the shop floor — physical goods you take home. Regulating is the maintenance crew — keeping the climate, water and pests in check behind the scenes. Supporting is the power station in the basement — it produces nothing you use directly, but every other department fails without it. Cultural is the experience the place gives you — the reason you'd miss it even if it sold you nothing.

THE EXAM-DEFINING INSIGHT

Why supporting services hold up everything else

Supporting services — soil formation, nutrient cycling and primary production (photosynthesis) — are different from the other three. You never consume them directly, they act over very long timescales, and they are the precondition for every other service. Damage the foundation and the whole structure comes down. Press the button and watch.

◆ NOTE

Write this in an exam “Supporting services such as nutrient cycling and soil formation are necessary for the production of all other ecosystem services. If they are degraded — for example by erosion stripping topsoil — provisioning (food, timber), regulating (water purification) and cultural (recreation) services all decline as a consequence, because the ecosystem can no longer sustain itself.”

THE HOOK THAT MAKES IT STICK

What if you had to pay for it?

Because ecosystem services arrive free, we under-value them — until they’re gone and we have to replace them with technology. These real cases show how expensive “free” really is, and they make brilliant exam evidence.

Service category	Free version	The replacement bill
Regulating	Forested watershed naturally purifies drinking water	New York City protected its Catskills watershed for ~US\$1.5 billion rather than build a filtration plant estimated at US\$6–8 billion plus running costs.
Regulating	Wild bees and insects pollinate crops	In parts of Sichuan, China, collapsed pollinator populations force farmers to pollinate pear blossom by hand with paintbrushes — slow, costly human labour.
Supporting	Decomposers and weathering build fertile soil	A few centimetres of topsoil take centuries to form. Once eroded, restoring productivity means years of fertiliser, irrigation and replanting.
Cultural	Reserves provide recreation & mental restoration	Hard to price — but nature-based tourism and the documented mental-health value of green space run to billions annually, and can’t simply be rebuilt elsewhere.

◆ NOTE

Link it back This is why economists try to assign a dollar value to ecosystem services: putting a number on the “free” work makes it visible in decisions about clearing, mining or development. Expect this idea again in Sustainable Development (cost–benefit analysis) and value systems (KK14).

PRACTISE THE SKILL EXAMINERS TEST

Sort the service into the right category

◇ ACTIVE RECALL

The single most common ecosystem-services question is “classify this example.” Drag each example into a department of nature’s utility company — or tap it, then tap a bin. Get instant feedback.

READ THE QUESTION PROPERLY

Command-word decoder

◆ COMMAND WORDS

Marks are lost when students describe a question that asked them to evaluate. Tap a command word to see exactly what it’s demanding of you in an ecosystem-services answer.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

STRUCTURE THAT SCORES

The P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

High-mark ecosystem-services answers follow a rhythm: make a Point, back it with Evidence (a named example), then Link it to human well-being or the question. Watch one well-built sentence. Name the specific service and which of the four categories it belongs to. Don't be vague — “provides things” earns nothing. Anchor it in a real or case-study example. A named species, place or mechanism shows you understand, not just memorise. Spell out the benefit to human health or well-being — that's the part of the dot point students forget.

EXAM PRACTICE · 22 MARKS

Five questions — try, then reveal model answers

★ PRACTICE & SELF-MARK

Attempt each one in your logbook first. Then reveal a weak and a strong response, the marking guide, and award yourself a mark. Your running score sits in the dock, bottom-right.

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 Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.